

STUDENT MANAGER APP — BACKEND DATA MODEL

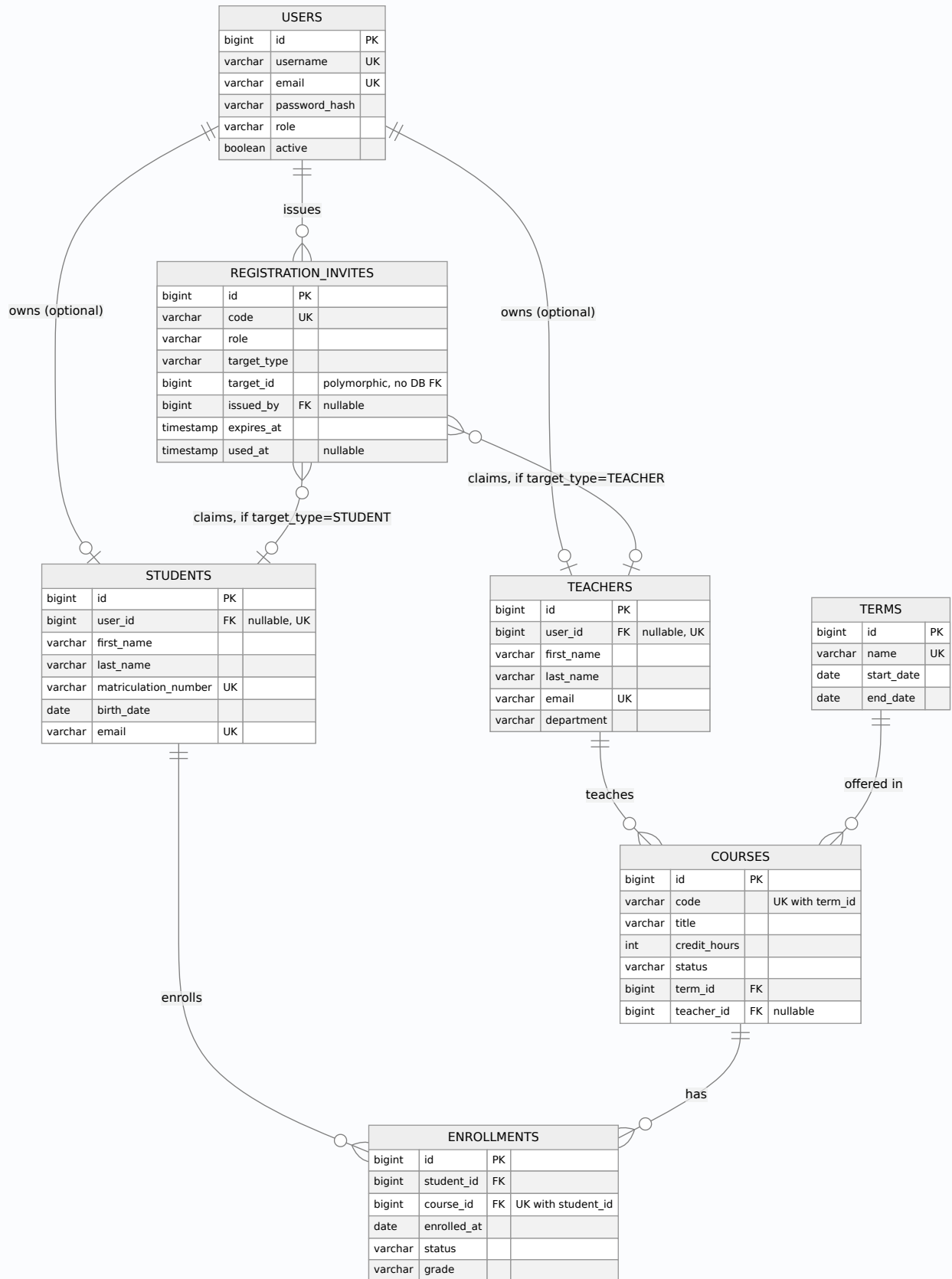
Database Design

Full relational schema for the student/course/enrollment domain, covering the accounts (users) & profile tables (students, teachers), the academic catalog (terms, courses), and the enrollments join table with grading and status tracking.

PostgreSQL · 6 tables · sample data shown is fictional, for illustration only

Entity Relationship Overview

Solid lines are foreign keys. `users.id` is optionally referenced by `students.user_id` and `teachers.user_id` — this replaces the old email-matching hack with a real, unique 1:1 link between a login account and its academic profile. Courses now belong to a `term` and are owned by a `teacher`.



What changed from the current schema

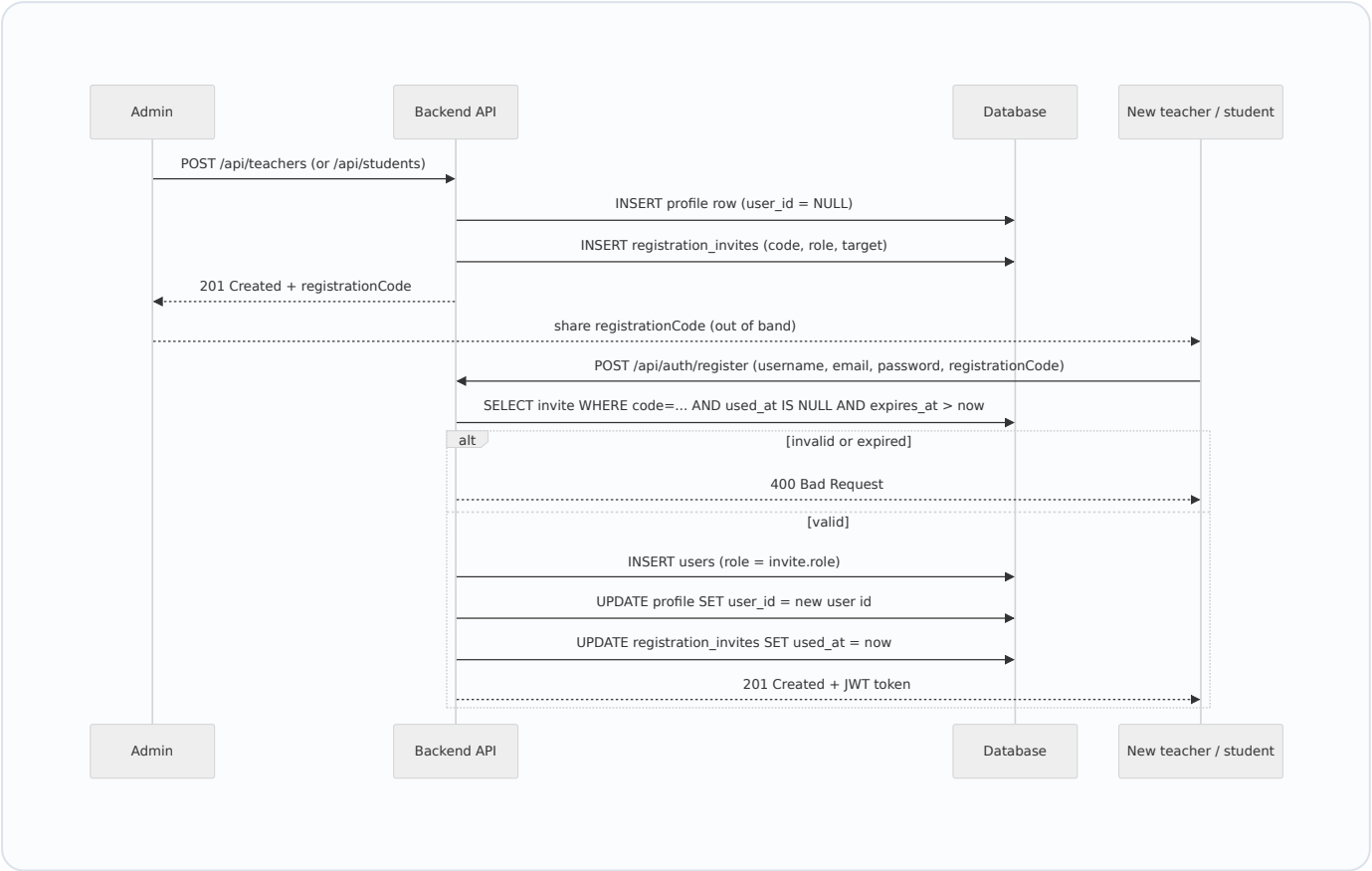
Why user_id is nullable

Added teachers, terms and registration_invites; added nullable unique user_id FKs on students/ teachers; courses.code is now unique per term instead of globally, so a course can be re-offered each term.

A student or teacher profile can exist before a login account is created (e.g. an admin pre-registers a new hire). The unique constraint still guarantees at most one profile per account.

Registration & Invite Flow

Replaces the old email-matching hack. An admin creates the profile row first (`user_id = NULL`) and gets back a one-time code; the person claims it through the existing public `/api/auth/register` endpoint. The role is always taken from the invite, never from the client — the same anti-privilege-escalation rule the codebase already applies to student self-registration, now generalized to cover teachers too.



1. users

Login identity and role only — no academic data lives here.

Column	Type	Constraints	Notes
id	bigint	PK, identity	
username	varchar(50)	NOT NULL, UNIQUE	login handle
email	varchar(255)	NOT NULL, UNIQUE	
password_hash	varchar(255)	NOT NULL	bcrypt
role	varchar(20)	NOT NULL	STUDENT / TEACHER / ADMIN
active	boolean	NOT NULL, default true	
created_at	timestamp	NOT NULL	
updated_at	timestamp	NOT NULL	

id	username	email	password_hash	role	active
1	admin.rossi	admin.rossi@campus.edu	\$2a\$10\$N9qo8u...Zo1e	ADMIN	true
2	j.keller	j.keller@campus.edu	\$2a\$10\$7xVs2p...q9Rk	TEACHER	true
3	m.duarte	m.duarte@campus.edu	\$2a\$10\$Lm3fA1...T4vn	TEACHER	true
4	s.omar	s.omar@campus.edu	\$2a\$10\$Qk91zR...Xp2c	TEACHER	true
5	lena.hoffmann	lena.hoffmann@student.campus.edu	\$2a\$10\$Bh6tY0...Wm8s	STUDENT	true
6	tom.becker	tom.becker@student.campus.edu	\$2a\$10\$Ux3jK9...Da7f	STUDENT	true
7	amara.diallo	amara.diallo@student.campus.edu	\$2a\$10\$Vp0nC2...Ye5g	STUDENT	true
8	yusuf.aydin	yusuf.aydin@student.campus.edu	\$2a\$10\$Rt8mZ4...Ju3k	STUDENT	true
9	chloe.martin	chloe.martin@student.campus.edu	\$2a\$10\$Fd2wQ7...Nb6h	STUDENT	true
10	noah.svensson	noah.svensson@student.campus.edu	\$2a\$10\$Ge9xL1...Kc4d	STUDENT	false

2. students

Academic profile. `user_id` is the fix for the old email-matching ownership check — row 7 shows a student with no linked account yet.

Column	Type	Constraints	Notes
<code>id</code>	bigint	PK, identity	
<code>user_id</code>	bigint	FK → users.id, UNIQUE, NULLABLE	one account per student, or none yet
<code>first_name</code>	varchar(100)	NOT NULL	
<code>last_name</code>	varchar(100)	NOT NULL	
<code>matriculation_number</code>	varchar(30)	NOT NULL, UNIQUE	
<code>birth_date</code>	date		
<code>email</code>	varchar(255)	NOT NULL, UNIQUE	contact only, no longer used for ownership
<code>created_at</code>	timestamp	NOT NULL	
<code>updated_at</code>	timestamp	NOT NULL	

id	user_id	first_name	last_name	matriculation_number	birth_date	email
1	5	Lena	Hoffmann	MAT-2025-0091	2003-04-12	lena.hoffmann@student.campus.edu
2	6	Tom	Becker	MAT-2025-0092	2002-11-30	tom.becker@student.campus.edu
3	7	Amara	Diallo	MAT-2024-0043	2001-07-08	amara.diallo@student.campus.edu
4	8	Yusuf	Aydin	MAT-2024-0044	2002-01-22	yusuf.aydin@student.campus.edu
5	9	Chloe	Martin	MAT-2025-0093	2003-09-15	chloe.martin@student.campus.edu
6	10	Noah	Svensson	MAT-2023-0011	2000-03-05	noah.svensson@student.campus.edu
7	NULL	Priya	Nair	MAT-2025-0094	2003-12-01	priya.nair@student.campus.edu

3. teachers

New table — previously TEACHER was only a role on users with no profile or course ownership.

Column	Type	Constraints	Notes
id	bigint	PK, identity	
user_id	bigint	FK → users.id, UNIQUE, NULLABLE	
first_name	varchar(100)	NOT NULL	
last_name	varchar(100)	NOT NULL	
email	varchar(255)	NOT NULL, UNIQUE	
department	varchar(100)		
created_at	timestamp	NOT NULL	
updated_at	timestamp	NOT NULL	

id	user_id	first_name	last_name	email	department
1	2	Jonas	Keller	j.keller@campus.edu	Computer Science
2	3	Marta	Duarte	m.duarte@campus.edu	Mathematics
3	4	Samira	Omar	s.omar@campus.edu	Physics
4	NULL	Elena	Petrov	e.petrov@campus.edu	Chemistry

4. registration_invites

target_type + target_id is a polymorphic reference (points into students or teachers depending on the value) — enforced in application code, not by a database FK. Row 5 is a pending invite (Priya Nair, not yet claimed); row 6 shows an expired, never-claimed invite for a new hire — the admin would need to reissue it.

Column	Type	Constraints	Notes
id	bigint	PK, identity	
code	varchar(40)	NOT NULL, UNIQUE	random token
role	varchar(20)	NOT NULL	STUDENT / TEACHER — assigned by admin, never by the claimant
target_type	varchar(20)	NOT NULL	STUDENT / TEACHER
target_id	bigint	NOT NULL	id in the students or teachers table
issued_by	bigint	FK → users.id, NULLABLE	admin who created it
expires_at	timestamp	NOT NULL	
used_at	timestamp	NULLABLE	set once claimed
created_at	timestamp	NOT NULL	

id	code	role	target_type	target_id	issued_by	expires_at	used_at
1	8f2a1c9e4b7d4f0a	TEACHER	TEACHER	1 (Keller)	1	2025-08-15	2025-08-05
2	3d7b6e21aa9c48f1	TEACHER	TEACHER	2 (Duarte)	1	2025-08-15	2025-08-06
3	c15f0e88b3a2440d	TEACHER	TEACHER	3 (Omar)	1	2025-08-15	2025-08-07
4	aa41d9c6e5f8412b	STUDENT	STUDENT	6 (Svensson)	1	2025-09-10	2025-09-01
5	f30b8d54c9e6471a	STUDENT	STUDENT	7 (Nair)	1	2026-09-30	NULL
6	9c04a71fe2d8463a	TEACHER	TEACHER	4 (Petrov)	1	2026-08-01	NULL

5. terms

New table — scopes courses and enrollments to an academic period.

Column	Type	Constraints	Notes
id	bigint	PK, identity	
name	varchar(50)	NOT NULL, UNIQUE	e.g. "Fall 2025"
start_date	date	NOT NULL	
end_date	date	NOT NULL	
created_at	timestamp	NOT NULL	
updated_at	timestamp	NOT NULL	

id	name	start_date	end_date
1	Fall 2025	2025-09-01	2025-12-19
2	Spring 2026	2026-02-02	2026-05-29

6. courses

code is unique per term_id (not globally) — row 1 and row 4 both use CS101, offered in two different terms by the same teacher.

Column	Type	Constraints	Notes
id	bigint	PK, identity	
code	varchar(20)	NOT NULL, UNIQUE with term_id	
title	varchar(150)	NOT NULL	
description	text		
credit_hours	int	NOT NULL	
status	varchar(20)	NOT NULL, default ACTIVE	ACTIVE / INACTIVE / ARCHIVED
term_id	bigint	FK → terms.id, NOT NULL	
teacher_id	bigint	FK → teachers.id, NULLABLE	unassigned until staffed
created_at	timestamp	NOT NULL	
updated_at	timestamp	NOT NULL	

id	code	title	credit_hours	status	term_id	teacher_id
1	CS101	Intro to Programming	5	ACTIVE	1 (Fall 2025)	1 (Keller)
2	MATH201	Linear Algebra	4	ACTIVE	1 (Fall 2025)	2 (Duarte)
3	PHYS150	Classical Mechanics	4	ACTIVE	1 (Fall 2025)	3 (Omar)
4	CS101	Intro to Programming	5	ACTIVE	2 (Spring 2026)	1 (Keller)
5	CS220	Data Structures	5	ACTIVE	2 (Spring 2026)	1 (Keller)
6	MATH305	Statistics	4	INACTIVE	2 (Spring 2026)	NULL

7. enrollments

Join table between `students` and `courses`, unique per pair. Status and grade are independent — a `CANCELLED` row can (and should) stay `NOT_GRADED`.

Column	Type	Constraints	Notes
<code>id</code>	<code>bigint</code>	PK, identity	
<code>student_id</code>	<code>bigint</code>	FK → <code>students.id</code> , NOT NULL	
<code>course_id</code>	<code>bigint</code>	FK → <code>courses.id</code> , NOT NULL, UNIQUE with <code>student_id</code>	
<code>enrolled_at</code>	<code>date</code>	NOT NULL	
<code>status</code>	<code>varchar(20)</code>	NOT NULL, default <code>PENDING</code>	<code>PENDING</code> / <code>CONFIRMED</code> / <code>CANCELLED</code>
<code>grade</code>	<code>varchar(15)</code>	NOT NULL, default <code>NOT_GRADED</code>	<code>A</code> / <code>B</code> / <code>C</code> / <code>D</code> / <code>F</code> / <code>NOT_GRADED</code>
<code>created_at</code>	<code>timestamp</code>	NOT NULL	
<code>updated_at</code>	<code>timestamp</code>	NOT NULL	

<code>id</code>	<code>student_id</code>	<code>course_id</code>	<code>enrolled_at</code>	<code>status</code>	<code>grade</code>
1	1 (Hoffmann)	1 (CS101 F25)	2025-08-20	CONFIRMED	A
2	1 (Hoffmann)	2 (MATH201 F25)	2025-08-20	CONFIRMED	B
3	2 (Becker)	1 (CS101 F25)	2025-08-21	CONFIRMED	C
4	2 (Becker)	3 (PHYS150 F25)	2025-08-22	CANCELLED	NOT_GRADED
5	3 (Diallo)	2 (MATH201 F25)	2025-08-19	CONFIRMED	A
6	3 (Diallo)	3 (PHYS150 F25)	2025-08-19	CONFIRMED	B
7	4 (Aydin)	1 (CS101 F25)	2025-08-25	PENDING	NOT_GRADED
8	5 (Martin)	3 (PHYS150 F25)	2025-08-23	CONFIRMED	D
9	6 (Svensson)	4 (CS101 S26)	2026-01-15	PENDING	NOT_GRADED
10	1 (Hoffmann)	5 (CS220 S26)	2026-01-10	CONFIRMED	NOT_GRADED
11	7 (Nair)	4 (CS101 S26)	2026-01-12	CONFIRMED	NOT_GRADED
12	4 (Aydin)	5 (CS220 S26)	2026-01-14	CANCELLED	NOT_GRADED